Course creation
Instructor.createCourse creates a Course in DRAFT status, assigns the current Instructor as its owner, and rejects a Specialty that the Instructor does not have.

Mandatory references
Course.specialty and Course.instructor are mandatory.
Enrollment.student and Enrollment.course are mandatory.

Enrollment rules
Student.applyCourse(course) uses the current system date. Enrollment is allowed only if the Course is PUBLISHED, the current date is before course.startDate, the Course has available capacity, and the Student is not already enrolled in that Course.
A Student may have at most one Enrollment per Course. Database constraint: unique {student_id, course_id}.

Pricing and approvals
The first enrollment by a Student for a given Specialty in a calendar month is charged at the base price. Further enrollments by the same Student for the same Specialty in the same month and year receive a 25% discount.
Enrollment.finalAmount stores the final amount charged.
ApprovalRequest.course is mandatory. ApprovalRequest.administrator is optional until an Administrator resolves the request. instructorComment and administratorComment are separate fields written by different roles.

Collection encapsulation
Student.enrollments, Course.enrollments, Instructor.courses, Instructor.specialties, Specialty.instructors, and Specialty.courses are exposed as read-only collection views. Callers cannot modify them directly.

«abstract»
User

id: Long
firstName: String
lastName: String
email: String {unique}
password: String

Student

- dni: String {unique}
- enrollments: Set<Enrollment>

+ applyCourse(course: Course): Enrollment
~ applyCourse(course: Course, enrollmentDate: LocalDate): Enrollment

Instructor

- specialties: Set<Specialty>
- courses: Set<Course>

+ createCourse(title: String, description: String, basePrice: BigDecimal, startDate: LocalDate, endDate: LocalDate, maxCapacity: Integer, specialty: Specialty): Course
+ addSpecialty(specialty: Specialty): void
+ removeSpecialty(specialty: Specialty): void

Administrator

Specialty

- id: Long
- name: String {unique}
- instructors: Set<Instructor>
- courses: Set<Course>

Course

- id: Long
- title: String
- description: String
- basePrice: BigDecimal {precision=12, scale=2}
- startDate: LocalDate
- endDate: LocalDate
- maxCapacity: Integer
- status: CourseStatus = DRAFT
- specialty: Specialty
- instructor: Instructor
- enrollments: Set<Enrollment>

+ hasAvailableCapacity(): boolean

ApprovalRequest

- id: Long
- requestDate: LocalDateTime
- resolutionDate: LocalDateTime [0..1]
- status: ApprovalStatus = SUBMITTED
- instructorComment: String [0..1]
- administratorComment: String [0..1]
- course: Course
- administrator: Administrator [0..1]

+ ApprovalRequest(course: Course)

«enumeration»
CourseStatus

DRAFT
PENDING_APPROVAL
PUBLISHED

«enumeration»
ApprovalStatus

SUBMITTED
APPROVED
REJECTED

Enrollment

- id: Long
- enrollmentDate: LocalDateTime
- finalAmount: BigDecimal {precision=12, scale=2}
- student: Student
- course: Course

«exception»
CourseEnrollmentException

+ CourseEnrollmentException(message: String)

0..*
has specialties
0..*
1
creates / owns
1
0..1
resolves
receives approval requests
1
0..*
classifies
1
0..*
0..*
has enrollments
1
0..*
enrolls in
1